From Memory to Causal History: Evidence-Bounded Retrodiction for Persistent Agents

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Abstract

Persistent-agent memory is usually evaluated as retention, retrieval, updating, and reuse of prior experience. We argue that a consequential long-term agent also needs to distinguish three questions that ordinary memory systems can conflate: what an actor was contemporaneously documented to know, what later evidence supports that the actor had access to at the time, and what is known retrospectively about the causal history that produced a present state. We propose **Evidence-Bounded Retrodiction (EBR)**, a position-and-prototype protocol that keeps these evidence views non-interchangeable. EBR combines provenance-aware temporal memory with actor-access assertions, mechanism-gated candidate histories, contradiction-preserving retrieval, and an explicit reserve for unenumerated explanations. EBR does not collapse evidence quality into an unvalidated scalar or normalize candidate histories into pseudo-probabilities. A small executable synthetic prototype demonstrates why event time and database record time are insufficient proxies for historical epistemic state: a valid-time-only baseline leaks post-event evidence, while a record-time-only baseline misses evidence that an actor saw before the decision but the system archived later. We position EBR as an integration contract for persistent agents rather than a new theory of causality, bitemporal data, provenance, or abduction, and specify a falsifiable evaluation agenda for temporal leakage, actor-state fidelity, causal-path recovery, contradiction handling, and abstention.

1. Position: memory needs epistemic time

What if an agent could remember not only what was decided, but reconstruct how the decision became possible - and then use the same machinery to ask of a present observation: what sequence of causes, choices, perturbations, and information flows could have produced it? The present is evidence left by a path through time, but retrospective evidence is not the same thing as information that was available to an actor along that path.

This distinction matters for persistent agents that both (i) reconstruct prior causes and (ii) learn from past decisions. *Retrodiction* is explicitly retrospective: later measurements, archives, and outcomes may be legitimate evidence about what happened. *Ex-ante appraisal*, by contrast, asks what could reasonably have been concluded at a historical decision point and must not silently import outcome knowledge. Hindsight bias changes judgments about what was foreseeable (Fischhoff, 1975), and outcome knowledge can distort evaluation of decision quality even when evaluators know the same pre-decision facts (Baron and Hershey, 1988). Recent work similarly shows that LLMs violate explicit temporal cutoffs by using post-cutoff knowledge (Liu et al., 2026).

We therefore advance a narrower position than "causal memory": persistent agents should keep distinct temporal-epistemic views so that later evidence may improve causal reconstruction without rewriting the historical state against which earlier reasoning is appraised. EBR instantiates this position as an auditable protocol and small prototype.

2. Related work and novelty boundary

Agent memory. Long-term agent memory already extends far beyond transcript storage. A-Mem dynamically organizes and evolves linked memories (Xu et al., 2025); LongMemEval evaluates temporal reasoning, knowledge updates, and abstention (Wu et al., 2025). Hindsight separates world, experience, observation, and opinion networks with temporal and graph retrieval (Latimer et al., 2026). APEX-MEM uses a temporal property graph and append-only storage (Banerjee et al., 2026), while MAGMA represents memory through semantic, temporal, causal, and entity graphs (Jiang et al., 2026). ReasoningBank distills reusable strategies from successful and failed trajectories (Ouyang et al., 2025). EBR does not claim these capabilities as new.

Temporal data and provenance. Valid time and transaction time are established temporal-database concepts (Jensen et al., 1994); bitemporal knowledge graphs have also combined these dimensions with uncertainty (Chekol and Stuckenschmidt, 2018). PROV-O provides a standard vocabulary for entities, activities, agents, and derivation (Lebo et al., 2013). EBR uses these foundations but adds a separately provenanced *actor-access* relation because neither when a fact was true nor when a database stored it determines when a particular actor could know it.

Backward explanation. Abduction and diagnosis have long generated explanations for observations (Kakas et al., 1992; Reiter, 1987; de Kleer and Williams, 1987). Bayesian accident reconstruction explicitly infers past event conditions from available evidence and supports counterfactual analysis (Davis, 2003). Zeno uses temporal provenance to diagnose upstream causes of present timing failures (Wu et al., 2019). Structural causal models provide the semantics required for intervention and counterfactual claims (Pearl, 2009). Thus, reconstructing causes backward from effects is not our novelty.

Residual contribution. The contribution is to make an explicit, executable protocol binding these traditions around a persistent-agent invariant: three evidence views are stored and queried as different objects, post-outcome appraisal cannot rewrite the contemporaneous view, and causal candidates are admitted through explicit mechanism bridges while unresolved and unmodeled alternatives remain visible. The paper therefore advances an architectural integration and evaluation target, not a claim of new causal identification theory or of priority over every possible combination of the underlying components.

3. Three-view temporal-epistemic protocol

Let an evidence item *x* carry an event/valid interval $I_v(x)$, system record time r(x), provenance pi(x), epistemic status sigma(x), and zero or more actor-access assertions A(a,x,t) indicating evidence that actor *a* had access to *x* by time *t*. Actor access is not inferred from valid time: a past event can remain known after its world-state interval ends. Nor is access inferred from record time: an actor may receive a message before a later system imports its archive.

For actor *a*, historical decision time t_i, and audit cutoff c >= t_i, define $C(a,x,t_i) = 1$ when access to x by t_i was contemporaneously attested by t_i, and $H_c(a,x,t_i) = 1$ when evidence available by cutoff c supports that historical access.

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K_att(a,t_i) = {x : r(x) <= t_i, C(a,x,t_i) = 1}
K_rec(a,t_i;c) = {x : r(x) <= c, H_c(a,x,t_i) = 1}
R(c) = {x : r(x) <= c} ∩ S
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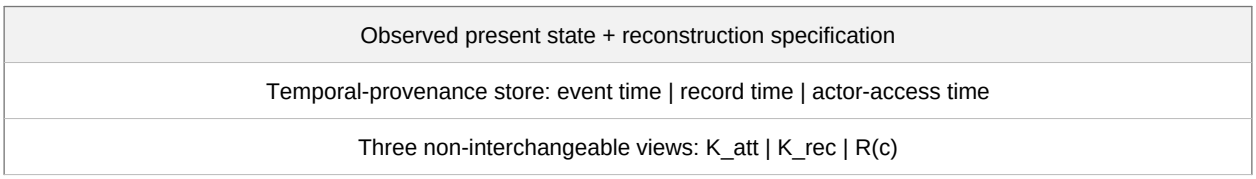
Here S is the declared retrieval scope. K_att is the contemporaneously attested epistemic record. K_rec is a later, provenance-tagged reconstruction of what the actor had accessed by t_i; it may contain archives ingested after the decision. It reconstructs historical access, not later validation of the claim's truth. Such validation remains retrospective evidence in R, which is available for causal reconstruction regardless of actor access.

Non-retroactivity invariant. Assume append-only transaction history: records and access attestations inserted after t_i cannot be back-written as if recorded earlier. Then later insertion cannot change K_att(a,t_i); it may only expand K_rec(a,t_i;c) or R(c). This is a direct use of established versioned/bitemporal semantics, not a new database theorem. Corrections are new records linked to the superseded assertion rather than destructive edits.

The distinction also clarifies "hindsight leakage." Retrodiction may use R(c) because its task is retrospective. An ex-ante appraisal of the actor's decision must use a declared K_att or K_rec snapshot and must hide outcome-conditioned retrieval/generation. Parametric post-cutoff knowledge in the underlying LLM remains a separate failure mode and should be audited with time-indexed controls such as ExAnte (Liu et al., 2026).

Figure 1. EBR architecture



Bounded candidate histories: retrieve | expand bridges | find contradictions | prune
Mechanism gate + evidence profile
Admissible frontier + unmodeled-history reserve
Determination OR abstention + discriminating evidence
Post-outcome appraisal -> reusable reasoning motifs for future cases only

EBR separates historical actor-state views from retrospective causal evidence. Later evidence can improve reconstruction and appraisal without mutating the contemporaneously attested state.

4. Evidence-bounded causal reconstruction

EBR begins with a Causal Reconstruction Specification: observed present state, temporal/geographic scope, high-confidence facts, assumptions, candidate hypotheses, exclusions, relevant actors, and a prompt for potentially missing variables or sources. User-supplied intuition is recorded as a provenance-tagged report or hypothesis; first-hand access claims and unsupported speculation are not treated equivalently.

A causal evidence matrix is an implementation view, not a new data model. Rows are candidate variables/events and columns are time slices; cells may represent observations, decisions, absences, contradictions, inferred states, or missing values. Each item retains source lineage, event time, record time, actor-access assertions, epistemic status, domain/resolution, and observer/selection metadata.

4.1 Candidate generation without exhaustive search

Candidate histories are not assumed exhaustive. A practical generator performs: (1) retrieval within the declared scope; (2) candidate-cause proposal; (3) mechanism-bridge expansion; (4) contradiction search; (5) source-lineage deduplication; (6) branch clustering/merging; and (7) beam or graph-search pruning. For ex-ante appraisal, candidate generation is a blind first pass executed only over the declared historical epistemic horizon; retrospective evidence and outcome-conditioned appraisal enter a separate second pass. This separation makes hypothesis-space leakage auditable rather than silently importing future information.

4.2 Mechanism gate

Temporal adjacency and similarity can nominate a candidate but do not establish causation. A required edge u -> v is represented as a *mechanism bridge* carrying: temporal status, a named mechanism class, supporting provenance roots, an expected/testable intermediate implication, contradiction status, and an epistemic label (documented, domain-supported, inferred, speculative, contradicted, or unknown). A history with an unresolved required bridge may remain an investigatory hypothesis, but the system does not call it a complete admissible explanation. In short: *similarity identifies candidates; causality requires mechanism and evidence*. Temporal coincidence alone is not causal admissibility.

Where an explicit structural causal model is available, interventions and counterfactuals can be evaluated under that model (Pearl, 2009). Without the structural assumptions needed for identification, EBR does not convert narrative sensitivity or temporal ordering into causal effect estimates.

4.3 Evidence profiles, contradictions, and open-set uncertainty

EBR deliberately avoids a universal weighted scalar or softmax-normalized candidate set: correlated evidence can otherwise be counted repeatedly, free weights are domain-dependent, and closed-set normalization incorrectly assigns all support to an incomplete candidate set.

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Phi(h) = < m(h), n_pi(h), c(h), u(h) >
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Here m is supported/required mechanism-bridge coverage, n_pi is the number of independent provenance roots after lineage deduplication, c is the burden of adjudicated contradictions under the domain evidence policy, and u is the

number of unresolved required bridges. The prototype uses a transparent non-compensatory ordering only for display: histories with complete bridges and no adjudicated hard contradiction precede incomplete or hard-contradicted histories; weak or disputed discordant evidence remains visible rather than acting as an automatic veto. Otherwise ties or incomparable candidates may remain. No component is called a probability. An explicit unmodeled-history reserve remains active, so the interface can state "strongest among enumerated admissible candidates" or abstain entirely.

Observation is also not collapsed into one penalty. If measuring, investigating, or disclosing information changes subsequent world state, that action becomes a causal event node; research participation can alter behavior (McCambridge et al., 2014). If the issue is which cases or records entered the evidence set, it is represented as selection/provenance structure rather than physical causation (Bareinboim and Pearl, 2012).

5. Executable synthetic prototype

We implemented a small deterministic prototype to test the protocol contract rather than claim real-world causal validity. The synthetic case is an automated production cell that loses power at 10:01; an operator made a response decision at 10:05; the later audit occurs at 12:00.

Six evidence items create two deliberately difficult timing cases. First, a grid warning was seen by the operator at 09:57 but its message archive entered the system at 11:30. Second, an 11:20 post-event inspection reports damage pertaining to a 09:56 disturbance; the inspection was not available to the operator at 10:05. Three candidate histories are enumerated: external grid disturbance, internal overload, and manual shutdown.

Historical view	Leakage	Actor precision / recall
Valid-time-only	1	0.83 / 1.00
Record-time-only	0	1.00 / 0.80
Three-view K_rec	0	1.00 / 1.00

Table 1. Synthetic contract checks. Actor precision/recall uses known toy access labels. Append-only non-retroactivity also passes.

At audit time, the external-disturbance history has all three required mechanism bridges supported by four independent provenance roots and no contradiction. The overload history has an unresolved required bridge. The manual-shutdown history has supporting chat evidence but is contradicted by an authentication log. The prototype therefore reports the external-disturbance history as the strongest admissible candidate *in the enumerated set*, keeps the unmodeled-history reserve active, and recommends an independent grid-telemetry check as discriminating evidence. These results demonstrate implementation of the temporal-epistemic and mechanism contracts only; they do not validate causal accuracy across domains.

6. Evaluation agenda

A decisive evaluation should mix controlled synthetic worlds with adjudicated real cases. Synthetic structural models can vary confounding, missing evidence, delayed ingestion, actor-access records, observer interventions, and deliberately omitted true histories. Real incident datasets should be selected only where event timestamps, source provenance, and a credible adjudicated causal account exist; disputed historical narratives should not be treated as ground truth.

Primary metrics are: (i) **temporal leakage**, including post-cutoff evidence used in ex-ante appraisal; (ii) **actor-state fidelity**, precision/recall of reconstructed access against known access logs; (iii) **causal path/bridge recovery** where ground truth is defined; (iv) **contradiction recall** and provenance-root correctness; (v) **selective risk/abstention** when decisive evidence or the true history is withheld; and (vi) **historical-state invariance** under later evidence insertion. Calibration metrics such as Brier or log loss are appropriate only for future implementations that produce validated probabilities.

Useful baselines isolate different claims: long-context/RAG without historical filtering; record-time-only temporal memory; Hindsight/APEX-MEM-style structured temporal memory; MAGMA-style multi-graph retrieval; and model-based/abductive diagnosis when a domain model is available. Ablations remove actor-access assertions, the mechanism gate, contradiction retrieval, or the unmodeled reserve. The falsifiable question is whether the three-view contract reduces historical-state errors and hindsight contamination without degrading causal recovery or abstention behavior.

7. Learning and broader research program

Persistent agents may also reuse prior reasoning, but outcome knowledge must not become retrospective proof that a process was good. We therefore separate ex-ante process quality, realized outcome, causal attribution of the outcome to the process, and unforeseeable shocks before updating reusable reasoning patterns. A lucky success should not receive the same reinforcement as a well-supported process that causally contributed to success; a sound decision defeated by an unforeseeable shock should not be treated as incompetent merely because its outcome was poor. This directly addresses classical hindsight/outcome-bias concerns (Fischhoff, 1975; Baron and Hershey, 1988) while remaining compatible with reasoning-memory systems such as ReasoningBank (Ouyang et al., 2025). Later appraisal may affect future retrieval policy; it never rewrites K_{att}.

EBR is domain-agnostic only at the protocol level. Industrial failure, aviation, cyber incidents, supply chains, organizational memory, historical analysis, and scientific hypothesis generation each require domain-specific mechanism models, evidence standards, and validation. The longer-term metaphor is a computational "time machine" only in the restrained sense of evidence-bounded reconstruction of prior states and pathways; inaccessible history remains inaccessible.

8. Conclusion

Persistent memory should distinguish what was contemporaneously documented, what later evidence supports an actor had access to, and what is known retrospectively about the causal path that produced an observed state. EBR turns that distinction into an auditable protocol: immutable historical state, later actor-access reconstruction, retrospective causal evidence, mechanism-gated branches, contradiction preservation, and open-set uncertainty. The synthetic prototype shows why valid time and record time alone are insufficient proxies for actor knowledge. The broader research claim is deliberately testable: if the three-view separation does not reduce temporal leakage and historical-state error relative to simpler temporal-memory baselines, the protocol adds no empirical value.

Limitations

EBR cannot recover causes absent from both its evidence sources and candidate-generation process, and the unmodeled reserve identifies but does not solve that limitation. Actor-access assertions may be incomplete, wrong, or privacy-sensitive; later reconstruction of access is an evidential claim, not mind reading. The mechanism gate can still admit plausible but false mechanisms when domain models or records are weak. The toy prototype is synthetic and deterministic, so its perfect actor-state recall is a contract check rather than evidence of real-world performance. No cross-domain causal-accuracy claim is made. High-stakes uses require domain expert review and should not infer guilt, medical diagnosis, or sensitive psychological states from circumstantial patterns. User-supplied hypotheses may broaden search but can inject bias and must retain provenance and epistemic status. Selection processes can make surviving records unrepresentative, while interventions can alter the system being studied; modeling these processes does not reconstruct missing reality. Finally, LLM parametric knowledge may leak post-cutoff information even when external retrieval is correctly bounded, requiring separate leakage audits.

Declaration on Generative AI

Generative AI tools were used to support literature discovery, adversarial review, drafting and restructuring of text, mathematical/notation checking, prototype and figure preparation, and document production. The author directed the research questions and conceptual requirements and remains responsible for the scientific claims, code, citations, and final manuscript. AI-generated suggestions were not treated as evidence; important bibliographic and technical

claims were checked against primary or official sources before inclusion.

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